CORE–13: Artificial Intelligence (Unit–1)

Introduction to Artificial Intelligence: Artificial Intelligence (AI) is a branch of Science which deals with helping machines to find solutions to complex problems in a more human-like fashion. This generally involves borrowing characteristics from human intelligence, and applying them as algorithms in a computer friendly way.

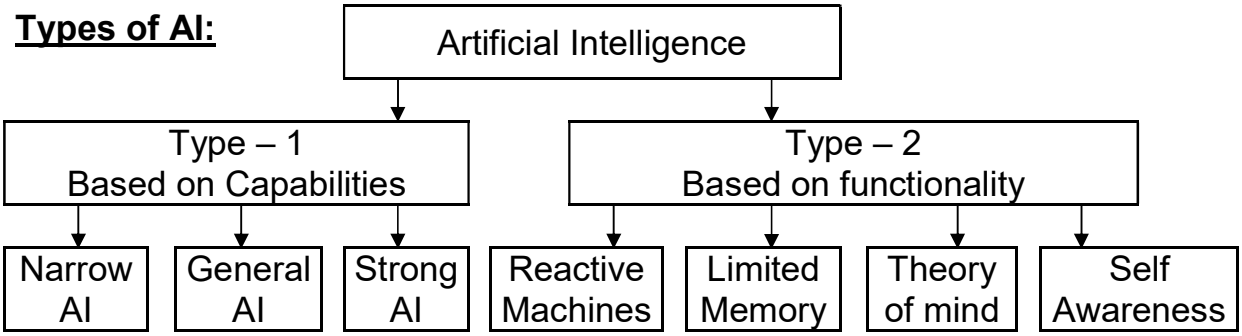
AI can be defined as: "AI is the science and engineering of making intelligent machines that can **think** like humans, **act** like humans, think rationally and can act rationally**".** It can be viewed from a variety of perspectives:

- From the perspective of intelligence, artificial intelligence is making machines "intelligent" -- acting as we would expect people to act.
  - The inability to distinguish computer responses from human responses is called the Turing test.
  - Intelligence requires knowledge
- From a business perspective AI is a set of very powerful tools, and methodologies for using those tools to solve business problems.
- From a programming perspective, AI includes the study of symbolic programming, problem solving, and search.

Components of Artificial Intelligence:

- **Reasoning, problem-solving:** Researchers had developed machines with algorithms that enable machines to solve puzzles or quiz similar to humans. AI can also deal with uncertain or incomplete information through advanced algorithms.
- **Knowledge Representation:** It is the representation of all the knowledge which is stored by an agent to make an expert system. Knowledge can be a set of objects, relations, concepts, or properties.
- **Planning:** Intelligent agents should be able to set goals and make plans to achieve those goals. They should be able to visualize the future and make predictions about their actions taken for achieving the goal.
- **Learning:** It is the study of the computer algorithms which improve automatically through experiences. This concept is known as Machine Learning.
- **Natural Language Processing:** This processing enables a machine to read and understand human language by processing the human language into machine language.
- **Perception:** An ability of the machine to use input from sensors, microphones, wireless signals, etc. for understanding different aspects of the world.

Types of AI:



AI type-1: Based on Capabilities

1. Weak AI or Narrow AI: Narrow AI is a type of AI which is able to perform a dedicated task with intelligence. The most common and currently available AI is Narrow AI in the world of Artificial Intelligence. Narrow AI cannot perform beyond its field or limitations, as it is only trained for one specific task. Hence it is also termed as weak AI. Narrow AI can fail in unpredictable ways if it goes beyond its limits. Some Examples of Narrow AI are **playing chess**, **purchasing suggestions on e-commerce site**, **self-driving cars**, **speech recognition**, and **image recognition**.

2. General AI: General AI is a type of intelligence which could perform any intellectual task with efficiency like a human. The idea behind the general AI is to make such a system which could be smarter and think like a human by its own. Currently, there is no such system exist which could come under general AI and can perform any task as perfect as a human. The worldwide researchers are now focusing on developing machines with General AI.

3. Super AI: Super AI is a level of Intelligence of Systems at which machines could surpass human intelligence, and **can perform any task better than human with cognitive properties.** It is an outcome of general AI. Some key characteristics of strong AI include capability include the **ability to think**, **to reason**, **solve the puzzle**, **make judgments**, **plan**, **learn**, and communicate by its own. Super AI is still a hypothetical concept of Artificial Intelligence. Development of such systems in real is a changing task.

Artificial Intelligence type-2: Based on functionality:

1. Reactive Machines: Purely reactive machines are the most basic types of Artificial Intelligence. Such AI systems **do not store past experiences** for future actions. These machines only focus on current scenarios and react on it as per possible best action. IBM's Deep Blue system and Google's AlphaGo are the examples of reactive machines.

2. Limited Memory: Limited memory machines **can store past experiences** or some data for a short period of time. These machines can use stored data for a limited time period only. Self-driving cars is one of the best examples of Limited Memory systems. These cars can store recent speed of nearby cars, the distance of other cars, speed limit, and other information to navigate the road.

3. Theory of Mind: Theory of Mind AI should understand the human emotions, people, beliefs, and be able to interact socially like humans. These types of AI machines are still not developed, but researchers are making lots of efforts and improvement for developing such AI machines.

4. Self-Awareness: Self-awareness AI is the future of Artificial Intelligence. These machines will be super intelligent, and will have their own consciousness, sentiments, and self-awareness. These machines will be smarter than human mind. Self-Awareness AI does not exist in reality yet and it is a hypothetical concept.

Background and Applications:

1. AI in Astronomy: Artificial Intelligence can be very useful to solve complex universe problems. AI technology can be helpful for understanding the universe such as how it works, origin, etc.

2. AI in Healthcare: In the last, five to ten years, AI becoming more advantageous for the healthcare industry and going to have a significant impact on this industry. Healthcare Industries are applying AI to make a better and faster diagnosis than humans. AI can help doctors with diagnoses and can inform when patients are worsening so that medical help can reach to the patient before hospitalization.

3. AI in Gaming: AI can be used for gaming purpose. The AI machines can play strategic games like chess, where the machine needs to think of a large number of possible places.

4. AI in Finance: AI and finance industries are the best matches for each other. The finance industry is implementing automation, chatbot, adaptive intelligence, algorithm trading, and machine learning into financial processes.

5. AI in Data Security: The security of data is crucial for every company and cyber-attacks are growing very rapidly in the digital world. AI can be used to make your data more safe and secure. Some examples such as AEG bot, AI2 Platform are used to determine software bug and cyber-attacks in a better way.

6. AI in Social Media: Social Media sites such as Facebook, Twitter, and Instagram contain billions of user profiles, which need to be stored and managed in a very efficient way. AI can organize and manage massive amounts of data. AI can analyze lots of data to identify the latest trends and requirement of different users.

7. AI in Travel &Transport: AI is becoming highly demanding for travel industries. AI is capable of doing various travel related works such as from making travel arrangement to suggesting the hotels, flights, and best routes to the customers. Travel industries are using AI-powered chatbots which can make human-like interaction with customers for better and fast response.

8. AI in Automotive Industry: Some Automotive industries are using AI to provide virtual assistant to their user for better performance. Such as Tesla has introduced TeslaBot, an intelligent virtual assistant. Various Industries are currently working for developing self-driven cars which can make your journey more safe and secure.

9. AI in Robotics: Artificial Intelligence has a remarkable role in Robotics. Usually, general robots are programmed such that they can perform some repetitive task, but with the help of AI, we can create intelligent robots which can perform tasks with their own experiences without pre-programmed. Humanoid Robots are best examples for AI in robotics, recently the intelligent Humanoid robot named as Erica and Sophia has been developed which can talk and behave like humans.

10. AI in Entertainment: We are currently using some AI based applications in our daily life with some entertainment services such as Netflix or Amazon. With the help of ML/AI algorithms, these services show the recommendations for programs or shows.

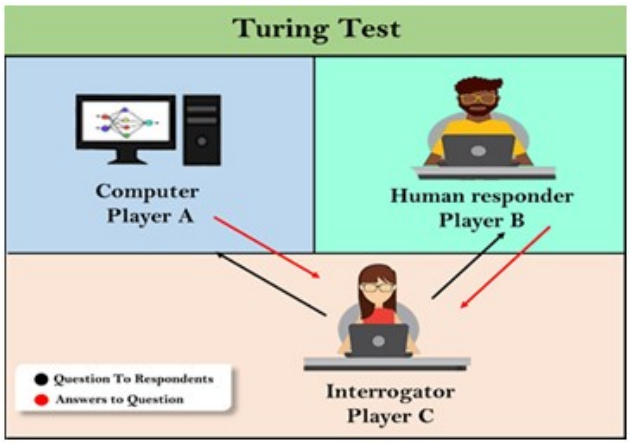
11. AI in Agriculture: Agriculture is an area which requires various resources, labor, money, and time for best result. Now a day's agriculture is becoming digital, and AI is emerging in this field. Agriculture is applying AI as agriculture robotics, solid and crop monitoring, predictive analysis. AI in agriculture can be very helpful for farmers.

12. AI in E-commerce: AI is providing a competitive edge to the e-commerce industry, and it is becoming more demanding in the e-commerce business. AI is helping shoppers to discover associated products with recommended size, colour, or even brand.

13. AI in education: AI can automate grading so that the tutor can have more time to teach. AI chatbot can communicate with students as a teaching assistant. AI in the future can be work as a personal virtual tutor for students, which will be accessible easily at any time and any place.

Turing Test Approach to AI (Acting humanly):

In 1950, Alan Turing introduced a test to check whether a machine can think like a human or not, this test is known as the Turing Test. In this test, Turing proposed that the computer can be said to be an intelligent if it can mimic human response under specific conditions which considered the question, "Can Machine think?"



The Turing test is based on a party game "Imitation game," with some modifications. This game involves three players in which one player is Computer, another player is human responder, and the third player is a **human Interrogator**, who is isolated from other two players and **his job is to find that which player is machine among two of them.**

Let us consider, Player A is a computer, Player B is human, and Player C is an interrogator. Interrogator is aware that one of them is machine, but he needs to identify this on the basis of questions and their responses.

The conversation between all players is via keyboard and screen so the result would not depend on the machine's ability to convert words as speech.

The test result does not depend on each correct answer, but only how closely its responses like a human answer. The computer is permitted to do everything possible to force a wrong identification by the interrogator.

In this game, if an interrogator would not be able to identify which is a machine and which is human, then the computer passes the test successfully, and the machine is said to be intelligent and can think like a human.

Features required for a machine to pass the Turing test:

- **Natural language processing:** NLP is required to communicate with Interrogator in general human language like English.
- **Knowledge representation:** To store and retrieve information during the test.
- **Automated reasoning:** To use the previously stored information for answering the questions.
- **Machine learning:** To adapt new changes and can detect generalized patterns.
- **Vision (For total Turing test):** To recognize the interrogator actions and other objects during a test.
- **Robotics / Motor Control (For total Turing test):** To act upon objects if requested.

Rational Agent approaches to AI (Acting rationally):

Rationality: Rationality is nothing but status of being reasonable, sensible, and having good sense of judgment. Rationality is concerned with expected actions and results depending upon what the agent has perceived. Performing actions with the aim of obtaining useful information is an important part of rationality.

Rational Agent: A rational agent is one that acts so as to achieve the best outcome or, when there is uncertainty, the best expected outcome.

Rationality of an agent depends on the following −

- The performance measures, which determine the degree of success.
- Agent's Percept Sequence till now.
- The agent's prior knowledge about the environment.
- The actions that the agent can carry out.

A rational agent always performs right action, where the right action means the action that causes the agent to be most successful in the given percept sequence.

Introduction to Intelligent Agents: An AI system can be defined as the study of the rational agent and its environment. The agents sense the environment through sensors and act on their environment through actuators. An AI agent can have mental properties such as knowledge, belief, intention, etc.

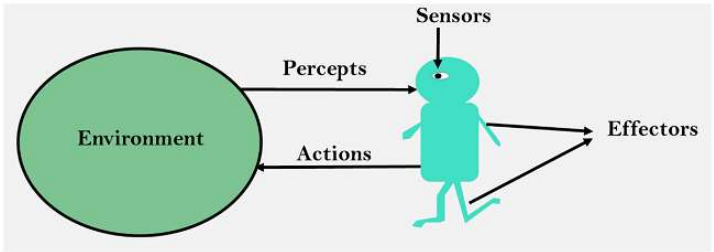
What is an Agent? - An agent can be anything that perceive its environment through sensors and act upon that environment through actuators. An Agent runs in the cycle of perceiving, thinking, and acting. An agent can be:

- **Human-Agent**: A human agent has eyes, ears, and other organs which work for sensors and hand, legs, vocal tract work for actuators.
- **Robotic Agent**: A robotic agent can have cameras, infrared range finder, NLP for sensors and various motors for actuators.
- **Software Agent**: Software agent can have keystrokes, file contents as sensory input and act on those inputs and display output on the screen.

Hence the world around us is full of agents such as thermostat, cell phone, camera, and even we are also agents. Before moving forward, we should first know about sensors, effectors, and actuators.

Sensor: Sensor is a device which detects the change in the environment and sends the information to other electronic devices. An agent observes its environment through sensors.

Effectors: Effectors are the devices which affect the environment. Effectors can be legs, wheels, arms, fingers, wings, fins, and display screen.



Actuators: Actuators are the component of machines that converts energy into motion. The actuators are only responsible for moving and controlling a system. An actuator can be an electric motor, gears, rails, etc.

Intelligent Agents: An intelligent agent is an autonomous entity which acts upon an environment using sensors and actuators for achieving goals. An intelligent agent may learn from the environment to achieve their goals. A thermostat is an example of an intelligent agent.

Following are the main four rules for an AI agent:

- Rule 1: An AI agent must have the **ability to perceive** the environment.
- Rule 2: The observation must be used to **make decisions**.
- Rule 3: Decision should **result in an action**.
- Rule 4: The action taken by an AI agent must be a **rational action**.

Structure of an AI Agent: The task of AI is to design an agent program which implements the agent function. The structure of an intelligent agent is a combination of architecture and agent program. It can be viewed as:

Agent = Architecture + Agent program

Following are the main three terms involved in the structure of an AI agent:
Architecture: Architecture is machinery that an AI agent executes on.
Agent Function: Agent function is used to map a percept to an action.

$f: P^* \rightarrow A$

Agent program: Agent program is an implementation of agent function. An agent program executes on the physical architecture to produce function f.

PEAS Representation: PEAS is a type of model on which an AI agent works upon. When we define an AI agent or rational agent, then we can group its properties under PEAS representation model. It is made up of four words:

- P: Performance measure
- E: Environment
- A: Actuators
- S: Sensors

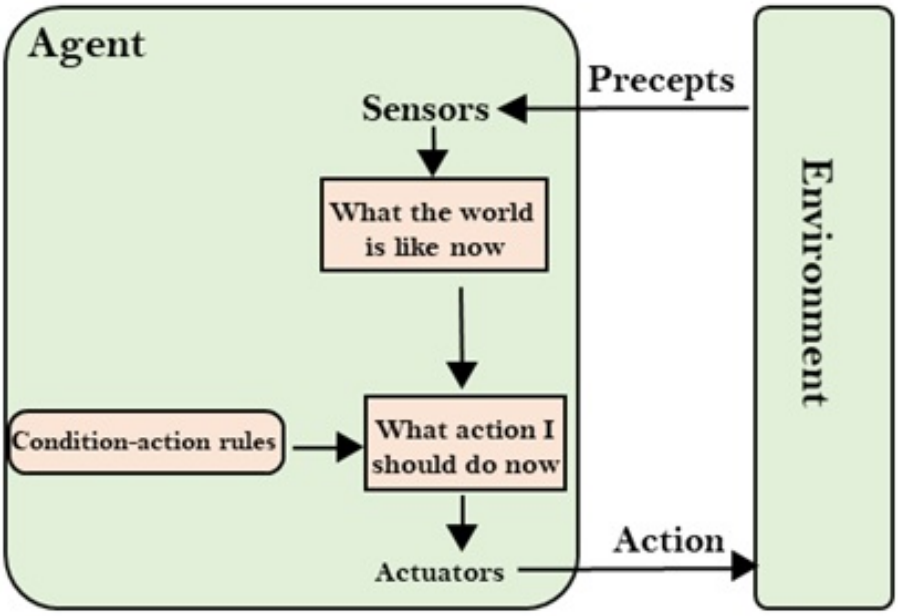
Example of Agents with their PEAS representation

Agent	Performance measure	Environment	Actuators	Sensors
self-driving cars	Safety, time, legal drive, comfort	Roads, other vehicles, road signs, pedestrian	Steering, accelerator, brake, signal, horn	Camera, GPS, speedometer, odometer, accelerometer, sonar.
Medical Diagnose	Healthy patient Minimized cost	Patient Hospital, Staff	Tests Treatments	Keyboard (Entry of symptoms)
Vacuum Cleaner	Cleanness Efficiency Battery life Security	Room, Table Wood floor Carpet Various obstacles	Wheels Brushes Vacuum Extractor	Camera, Cliff sensor Dirt detection sensor Bump Sensor Infrared Wall Sensor
Part - picking Robot	Percentage of parts in correct bins.	Conveyor belt with parts, Bins	Jointed Arms Hand	Camera Joint angle sensors.

Types of AI Agents: Agents can be grouped into five classes based on their degree of perceived intelligence and capability. All these agents can improve their performance and generate better action over the time. These are given below:

- Simple Reflex Agent
- Model-based reflex agent
- Goal-based agents
- Utility-based agent
- Learning agent

1. Simple Reflex agent-

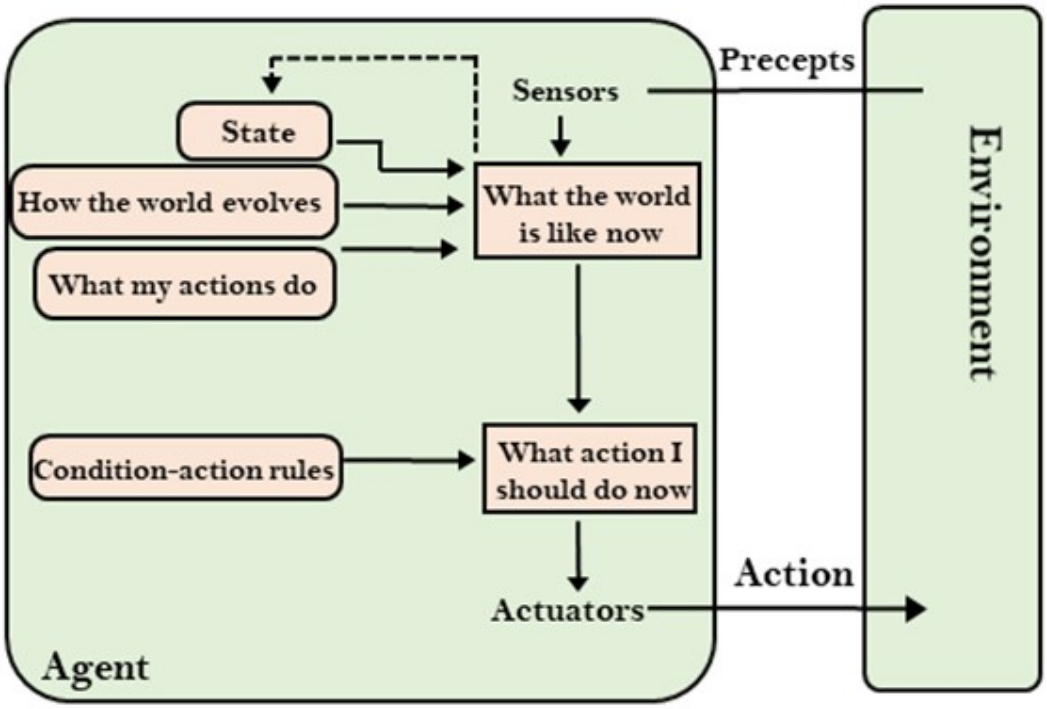


- The Simple reflex agents are the simplest agents. These agents take decisions on the basis of the current percept and ignore the rest of the percept history.
- These agents only succeed in the fully observable environment.
- The Simple reflex agents do not consider any part of percept history during their decision and action process.
- The Simple reflex agent works on Condition-action rule, which means it maps the current state to action. Such as a Room Cleaner agent, it works only if there is dirt in the room.
- Problems for the simple reflex agent design approach:
    - They have very limited intelligence.
    - They do not have knowledge of non-perceptual parts of the current state.
    - Mostly too big to generate and to store.
    - Not adaptive to changes in the environment.

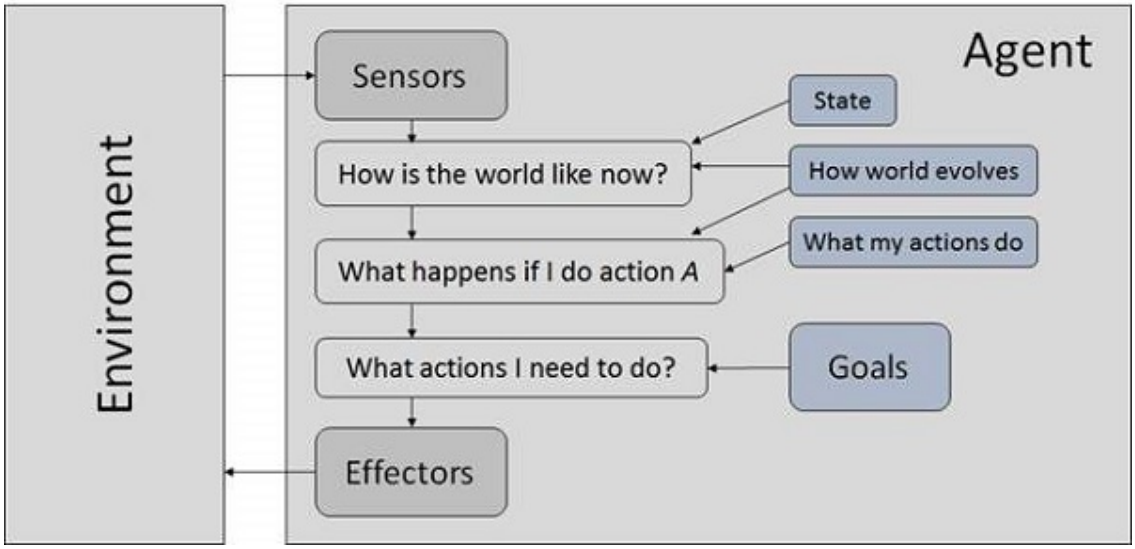
2. Model-based reflex agent-

- The Model-based agent can work in a partially observable environment, and track the situation.
- A model-based agent has two important factors:
    - **Model**: It is knowledge about "how things happen in the world," so it is called a Model-based agent.
    - **Internal State**: It is a representation of the current state based on percept history.
- These agents have the model, "which is knowledge of the world" and based on the model they perform actions.

- Updating the agent state requires information about:
    a. How the world evolves
    b. How the agent's action affects the world.



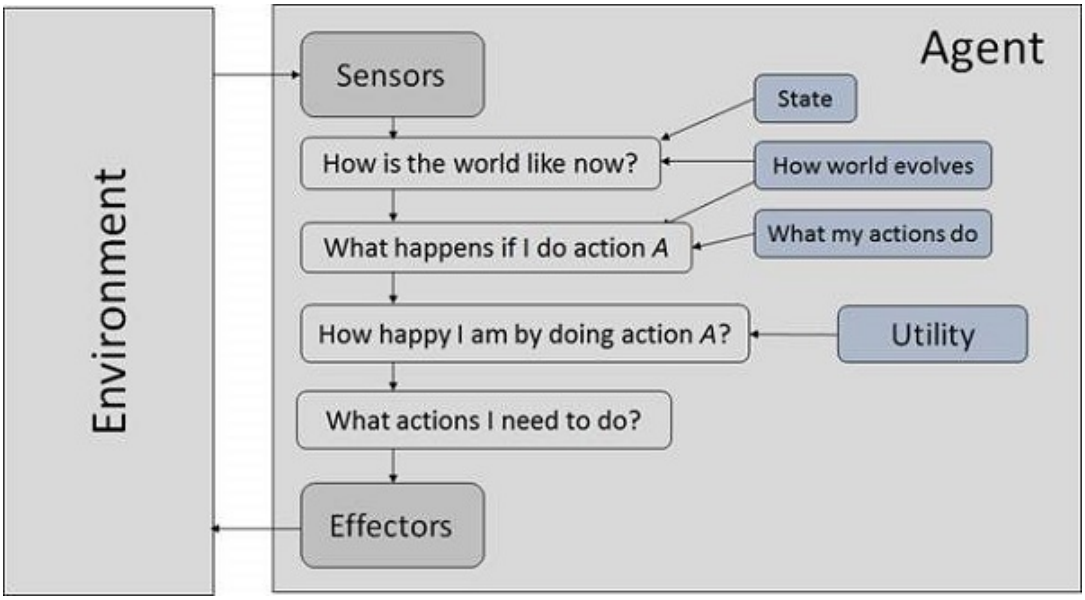
3. Goal-based agents-



- The knowledge of the current state environment is not always sufficient to decide for an agent to what to do.
- The agent needs to know its goal which describes desirable situations.
- Goal-based agents expand the capabilities of the model-based agent by having the "goal" information.
- They choose an action, so that they can achieve the goal.
- These agents may have to consider a long sequence of possible actions before deciding whether the goal is achieved or not. Such considerations of

different scenario are called searching and planning, which makes an agent proactive.

4. Utility-based agents-

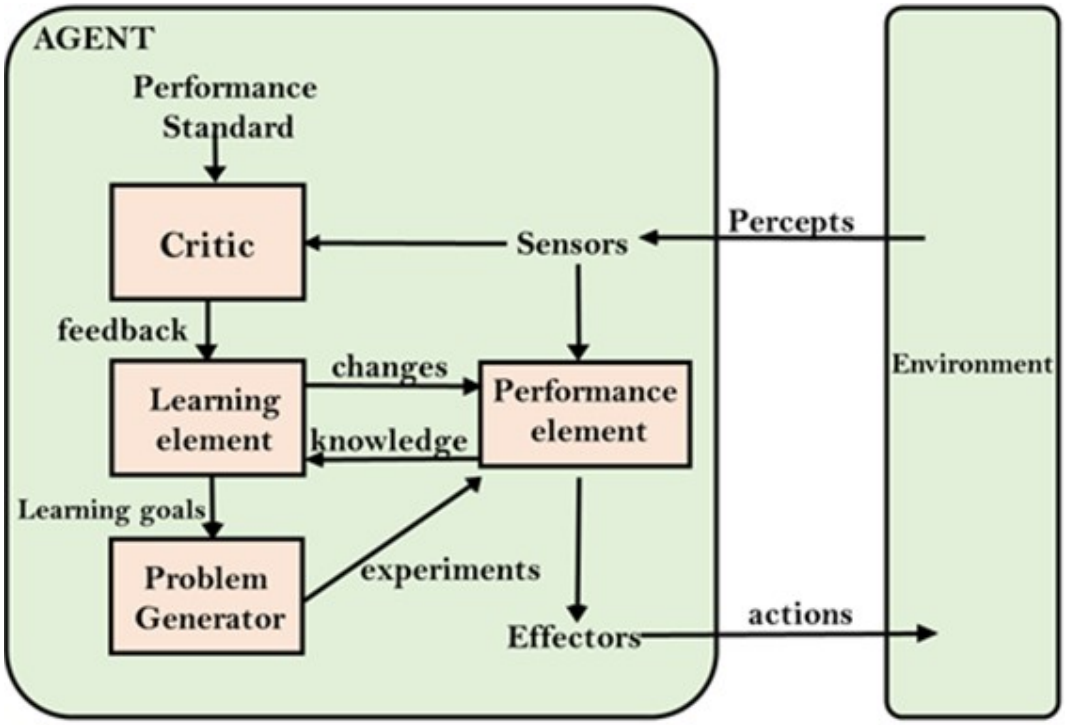


- These agents are similar to the goal-based agent but provide an extra component of utility measurement which makes them different by providing a measure of success at a given state.
- Utility-based agent act based not only goals but also the best **way to achieve the goal.**
- The Utility-based agent is useful when there are multiple possible alternatives, and an agent has to choose in order to perform the best action.
- The utility function maps each state to a real number to check how efficiently each action achieves the goals.

5. Learning Agents-

- A learning agent in AI is the type of agent which can learn from its past experiences, or it has **learning capabilities**.
- It starts to act with basic knowledge and then able to **act and adapt automatically through learning**.
- A learning agent has mainly four conceptual components, which are:
  a. **Learning element**: It is responsible for making improvements by learning from environment
  b. **Critic**: Learning element takes feedback from critic which describes that how well the agent is doing with respect to a fixed performance standard.
  c. **Performance element**: It is responsible for selecting external action.
  d. **Problem generator**: This component is responsible for suggesting actions that will lead to new and informative experiences.

Hence, learning agents are able to **learn, analyze performance, and look for new ways to improve the performance**.



Environment: An environment is everything in the world which surrounds the agent, but it is not a part of an agent itself. An environment can be described as a situation in which an agent is present. The environment is where agent lives, operate and provide the agent with something to sense and act upon it.

Properties of Environment:

1. Fully observable vs Partially Observable
2. Deterministic vs Stochastic
3. Episodic vs sequential
4. Single-agent vs Multi-agent
5. Static vs Dynamic
6. Discrete vs Continuous

1. Fully observable vs Partially Observable:

- If an agent sensor can sense or access the complete state of an environment at each point of time then it is a fully observable environment, else it is partially observable.
- A fully observable environment is easy as there is no need to maintain the internal state to keep track history of the world.
- An agent with no sensors in all environments then such an environment is called as unobservable.

2. Deterministic vs Stochastic:

- If an agent's current state and selected action can completely determine the next state of the environment, then such environment is called a deterministic environment.

- A stochastic environment is random in nature and cannot be determined completely by an agent.
- In a deterministic, fully observable environment, agent does not need to worry about uncertainty.

3. Episodic vs Sequential:

- In an episodic environment, there is a series of one-shot actions, and only the current percept is required for the action.
- However, in Sequential environment, an agent requires memory of past actions to determine the next best actions.

4. Single-agent vs Multi-agent

- If only one agent is involved in an environment, and operating by itself then such an environment is called single agent environment.
- However, if multiple agents are operating in an environment, then such an environment is called a multi-agent environment.

5. Static vs Dynamic:

- If the environment can change itself while an agent is deliberating then such environment is called a dynamic environment else it is called a static environment.
- Static environments are easy to deal because an agent does not need to continue looking at the world while deciding for an action.
- However for dynamic environment, agents need to keep looking at the world at each action.
- Taxi driving is an example of a dynamic environment whereas Crossword puzzles are an example of a static environment.

6. Discrete vs Continuous:

- If in an environment there are a **finite number of percept** and actions that can be performed within it, then such an environment is called a discrete environment else it is called continuous environment.
- A chess game comes under discrete environment as there are finite numbers of moves that can be performed.
- A self-driving car is an example of a continuous environment.